Hashtag Association Rule Mining

Which hashtags travel together? An Apriori market-basket analysis of 266,095 Big-Tech tweets, revealing the hidden co-occurrence patterns behind #hashtag chains.

Dataset	Bigtech_-_20-09-2020_till_13-10-2020.csv
Tweets analyzed	266,095
Tweets with 2+ hashtags	233,517 (88%)
Method	Apriori (min support 1%, min confidence 30%, lift > 1)
Rules discovered	170

Methodology

Each tweet was treated as a **transaction**, and the hashtags it contains as the **items** in that transaction — the same framework used for retail market-basket analysis ("customers who bought X also bought Y"). Hashtags were lower-cased and de-duplicated per tweet, and the 80 most frequent tags were retained as the mining vocabulary. The **Apriori algorithm** was then used to find frequent itemsets and derive association rules of the form **{A}** → **{B}**, meaning "tweets that use hashtag A also tend to use hashtag B."

Support	Confidence	Lift
% of all tweets containing **both** hashtags together.	P(consequent | antecedent) — of tweets with A, what % also had B.	How much more often A and B appear together than if they were independent. Lift > 1 = real association; higher = stronger.

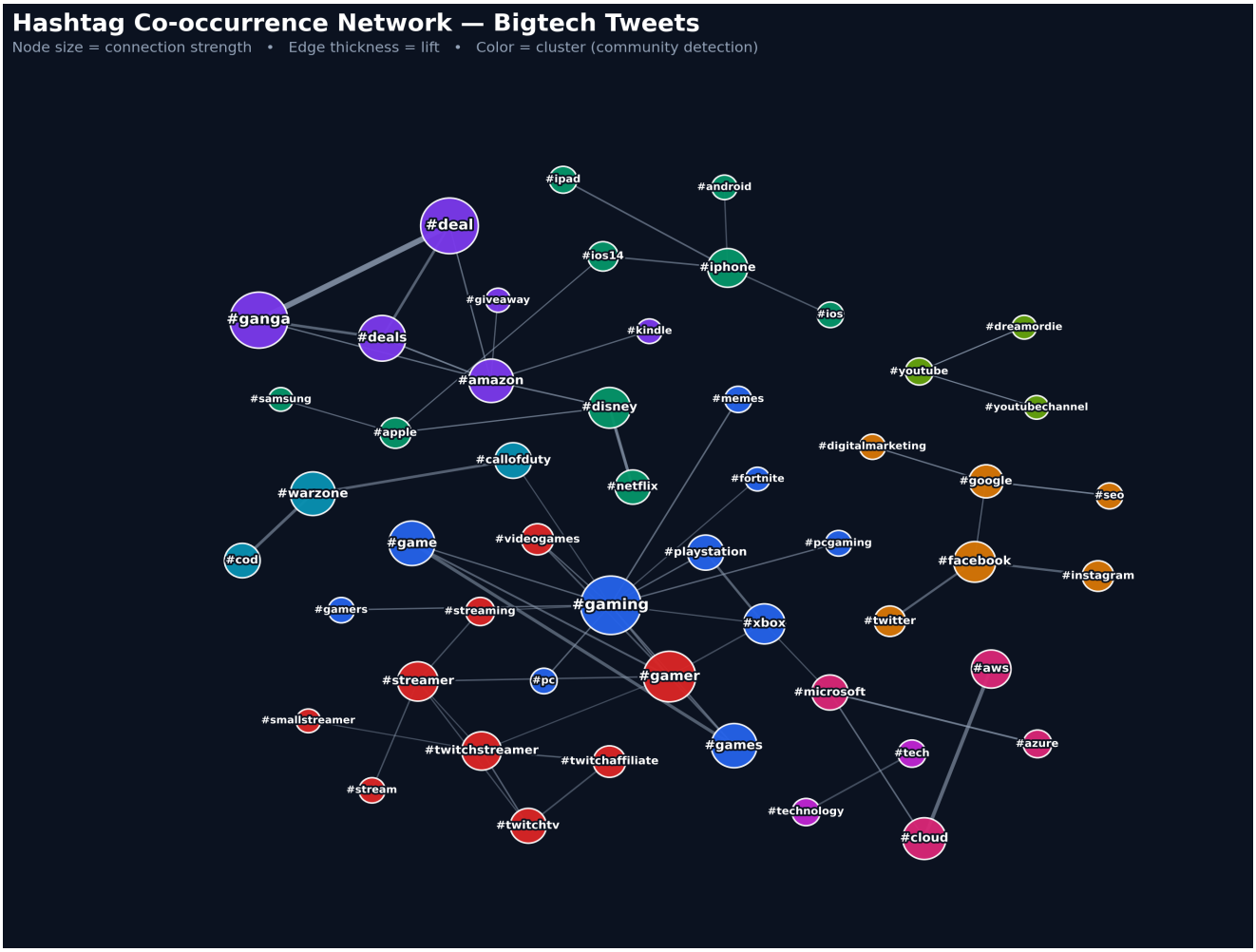
Parameters used: minimum support 1% (co-occurs in at least ~1,170 tweets), minimum confidence 30%, and lift > 1, keeping only itemsets up to size 3. Rules were de-duplicated so each hashtag pair/group is shown once, in the direction with the higher confidence.

At a glance

170	5	42.3×	100%
association rules found	thematic clusters	strongest lift	highest confidence

The Hashtag Network

Every node is a hashtag; an edge connects two hashtags that form a strong association rule. Node size reflects overall connection strength, edge thickness reflects lift, and color marks automatically-detected clusters (communities) of hashtags that travel together.



Gaming & Streaming

Twitch/YouTube streamer tags, console wars (Xbox vs PlayStation), and Call of Duty: Warzone chatter. ***(106 rules)***

Antecedent		Consequent	Support	Confidence	Lift
#game	→	*#games*	1.1%	50%	**20.8×**
#cod	→	*#warzone*	1.1%	58%	**18.5×**
#callofduty	→	*#warzone*	1.5%	52%	**16.6×**
#playstation	→	*#xbox*	1.0%	55%	**13.7×**
#game	→	*#gamer*	1.1%	51%	**10.2×**
#games	→	*#gamer*	1.1%	47%	**9.4×**
#gamer + #videogames	→	*#gaming*	1.1%	87%	**8.9×**
#gamer + #xbox	→	*#gaming*	1.0%	82%	**8.4×**
#videogames	→	*#gamer*	1.3%	40%	**8.1×**
#gamer + #youtube	→	*#gaming*	1.6%	74%	**7.6×**
#games + #twitch	→	*#gaming*	1.0%	73%	**7.5×**
#twitchtv	→	*#twitchaffiliate*	1.4%	34%	**7.4×**
#twitchtv	→	*#twitchstreamer*	2.1%	52%	**6.7×**
#gamer	→	*#streamer*	2.2%	43%	**6.6×**
#twitchaffiliate	→	*#twitchstreamer*	2.3%	50%	**6.5×**
#gamer + #streamer	→	*#gaming*	1.4%	62%	**6.4×**
#streamer + #youtube	→	*#gaming*	1.1%	61%	**6.3×**
#pcgaming	→	*#gaming*	1.0%	60%	**6.2×**
#xbox	→	*#gamer*	1.2%	31%	**6.1×**
#pc	→	*#gaming*	1.2%	59%	**6.1×**

+ 86 more rules in this cluster (see accompanying CSV for the full list).

Mobile & Big-Tech Ecosystem

Apple/Google/Android device tags and cross-platform mobile hashtags. ***(27 rules)***

Antecedent		Consequent	Support	Confidence	Lift
#amazon + #disney	→	*#netflix*	1.4%	99%	**19.9×**

Antecedent		Consequent	Support	Confidence	Lift
#apple + #disney	→	*#netflix*	1.3%	96%	**19.4×**
#disney	→	*#netflix*	1.6%	90%	**18.1×**
#technology	→	*#tech*	1.1%	30%	**8.1×**
#apple + #disney	→	*#amazon*	1.3%	95%	**7.8×**
#ipad	→	*#iphone*	1.1%	60%	**7.6×**
#apple + #netflix	→	*#amazon*	1.4%	88%	**7.2×**
#ios14	→	*#iphone*	1.3%	56%	**7.2×**
#seo	→	*#google*	2.2%	91%	**6.6×**
#disney	→	*#amazon*	1.4%	80%	**6.5×**
#ios	→	*#iphone*	1.8%	48%	**6.1×**
#digitalmarketing	→	*#google*	1.2%	79%	**5.7×**
#android	→	*#iphone*	1.5%	38%	**4.9×**
#disney	→	*#apple*	1.4%	79%	**4.7×**
#samsung	→	*#apple*	1.0%	65%	**3.9×**
#ios14	→	*#apple*	1.5%	64%	**3.8×**
#ios	→	*#apple*	2.3%	62%	**3.7×**
#ipad	→	*#apple*	1.1%	60%	**3.6×**
#amazon + #google	→	*#apple*	1.2%	60%	**3.6×**
#netflix	→	*#amazon*	1.8%	37%	**3.0×**

+ 7 more rules in this cluster (see accompanying CSV for the full list).

Social Media & Entertainment

Cross-posting handles (Instagram/Twitter/Facebook), YouTube and streaming-service tags. ***(22 rules)***

Antecedent		Consequent	Support	Confidence	Lift
#instagram	→	*#facebook*	1.0%	53%	**12.7×**
#twitter	→	*#facebook*	1.2%	52%	**12.4×**
#memes	→	*#gaming*	1.1%	67%	**6.9×**
#facebook	→	*#google*	2.4%	57%	**4.1×**
#fortnite	→	*#gaming*	1.0%	39%	**4.0×**

Antecedent		Consequent	Support	Confidence	Lift
#dreamordie	→	*#youtube*	5.0%	100%	**4.0×**
#youtubechannel	→	*#youtube*	1.5%	97%	**3.9×**
#youtuber	→	*#youtube*	3.3%	94%	**3.8×**
#subscribe	→	*#youtube*	1.7%	90%	**3.6×**
#video	→	*#youtube*	1.7%	77%	**3.1×**
#memes	→	*#youtube*	1.1%	72%	**2.9×**
#fortnite	→	*#twitch*	2.0%	76%	**2.5×**
#music	→	*#youtube*	2.0%	60%	**2.4×**
#spotify	→	*#apple*	1.1%	40%	**2.4×**
#instagram	→	*#youtube*	1.1%	57%	**2.3×**
#memes	→	*#twitch*	1.1%	70%	**2.3×**
#follow	→	*#youtube*	1.0%	52%	**2.1×**
#follow	→	*#twitch*	1.2%	61%	**2.0×**
#twitter	→	*#youtube*	1.1%	49%	**2.0×**
#spotify	→	*#youtube*	1.4%	48%	**1.9×**

+ 2 more rules in this cluster (see accompanying CSV for the full list).

E-commerce / Deals

A tight, near-automated cluster of deal-posting hashtags (very high lift — likely bot/affiliate chains). ***(12 rules)***

Antecedent		Consequent	Support	Confidence	Lift
#deals + #ganga	→	*#deal*	1.2%	100%	**42.3×**
#ganga	→	*#deal*	2.2%	99%	**41.7×**
#amazon + #ganga	→	*#deal*	1.5%	98%	**41.4×**
#ganga	→	*#deals*	1.2%	57%	**16.4×**
#deal	→	*#deals*	1.3%	54%	**15.6×**
#deals + #ganga	→	*#amazon*	1.2%	100%	**8.1×**
#deal + #deals	→	*#amazon*	1.3%	100%	**8.1×**
#deals	→	*#amazon*	3.1%	90%	**7.3×**
#ganga	→	*#amazon*	1.6%	72%	**5.8×**

Antecedent		Consequent	Support	Confidence	Lift
#deal	→	*#amazon*	1.6%	69%	**5.6×**
#kindle	→	*#amazon*	1.2%	61%	**5.0×**
#giveaway	→	*#amazon*	1.2%	53%	**4.4×**

Cloud & Enterprise Tech

AWS, Azure, Microsoft, and cloud-computing tags. ***(3 rules)***

Antecedent		Consequent	Support	Confidence	Lift
#aws	→	*#cloud*	1.1%	64%	**24.6×**
#azure	→	*#microsoft*	2.6%	88%	**8.8×**
#cloud	→	*#microsoft*	1.7%	67%	**6.7×**